

Reinforcement Learning

Programming Assignment - 2

REPORT

Team-04

Team Members:

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 - K Harpith Rao - SE22UCSE141
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Contributions:

- **K Sai Bharat Kumar Varma (SE22UCSE125)**
 - **Section 5.2:** Implemented the value iteration algorithm in `value_iteration.py`. Tuned convergence thresholds and ensured action space handling was robust. Conducted multiple simulation runs to test stability and correctness.
 - **Section 6.1 and 6.3:** Contributed to Q1 (computation time comparison) and Q3 (validation of Reason 1) Designed the plot showing $|x-z|$ vs transmission policy and interpreted how threshold-based transmission aligns with energy efficiency.
- **Dev M Bandhiya (SE22UCSE078)**
 - **Section 4:** Collaborated in formulating MDP for all three problems. Defined states, actions, and transition dynamics. Worked on the derivation of the Bellman optimality equations for passive-active phase modelling.
 - **Section 5.1:** Developed the policy evaluation module in `policy_evaluation.py` for evaluating greedy policies. Focused on correct reward calculation and ensuring stability under stochastic solar energy.
- **G Aditya Vardhan (SE22UCSE092)**

- **Section 4:** Took the lead in theoretical setup of the MDP. Structured problem modelling for dynamic battery levels and active/passive slot mechanisms.
 - **Section 6:** Contributed to Q2 (Comparative Analysis) and led analysis for Q5 (validation of Reason 3) by interpreting results of optimal policy under slot-limited conditions and documenting insights.
 - **K Harpith Rao (SE22UCSE141)**
 - **Section 5.3:** Implemented the policy iteration algorithm in `policy_iteration.py`. Integrated dynamic policy updates and state-specific action constraints. Ensured code modularity and proper convergence.
 - **Section 6:** Worked on Q4 (validation of Reason 2), analysing optimal behaviour under varying transmission reliability and during last active time slots. Supported graphical data analysis and conclusions.
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SECTION-4 (Theory Component):

4.1.

SECTION-4

PROBLEM-1

i) States and State Space (S):

→ The state at time t consists of:

- ϕ_t : current wind speed
- z_t : last successful transmission value stored at the monitoring station.

→ State space: $S = \{ \phi, z \mid \phi, z \in \{0, 0.05, \dots, 1\} \}$

here, S takes value from $S = \{0, 0.05, \dots, 1\}$ (normalized wind speed)

ii) Action and Action Space (A):

→ Action is the wind speed value WSN node decides to transmit.

→ Action space: $A : S = \{0, 0.05, \dots, 1\}$

iii) Transition probability:

→ Here, $P_{\phi, \phi'}$ denotes the probability of transitioning from ϕ to ϕ' .

• $\phi' = \phi$ w.p. $P_{\phi, \phi'}$

• $z' = \begin{cases} a & \text{w.p. } \lambda \\ z & \text{w.p. } 1-\lambda \end{cases}$

(iv) Reward function :

Goal is to minimize the squared error b/w the true wind speed and the last successful transmission.

$$R(\phi, z, a) = -(\phi - z')^2$$

$z' = a$ (transmission succeeds)
 $z' = z$ (transmission fails).

$$\Rightarrow R(\phi, z, a) = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2$$

(v) Bellman Optimality :

$$V^*(\phi, z) = \max_{a \in \mathcal{A}(\phi, z)} q^*(\phi, z, a)$$

$$\rightarrow \text{Here, } q^* = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2$$

$$+ \beta \sum_{\phi' \in \mathcal{S}} (\lambda p_{\phi, \phi'} V^*(\phi', a) + (1-\lambda) p_{\phi, \phi'} V^*(\phi', z))$$

4.2.

* Problem-2

(i) State Space (S) :

$$\mathcal{S} = \{(\phi, z, b) \mid \phi, z \in \mathcal{S}, b \in \{0, 1, \dots, B\}\}$$

ϕ : current wind speed at WSN node

z : last successful transmission value at monitoring station

b : battery charge ($b \in \{0, 1, \dots, B\}$)

(ii) Action and Action space (A):

At each time step, WSN must decide:

- whether to transmit (T):

$T=0 \rightarrow$ Do not transmit

$T=1 \rightarrow$ Transmit

- what value to transmit ($T=1$):

value must be from $S = \{0, 0.05, \dots, 1\}$

$\rightarrow$ Action space: $A = \{(T, a) \mid T \in \{0, 1\}, a \in S\}$

where, if $T=0 \rightarrow$ no transmission

if $T=1 \rightarrow$ transmits but iff $b \geq n$

(iii) Transition probability:

- $\phi' = \phi$ w.p. P_ϕ, ϕ'

- $z' = \begin{cases} a & \text{w.p. } \lambda \\ z & \text{w.p. } 1-\lambda \end{cases}$

solar energy distribution
 $\in (0, 1, \dots, \Delta)$

- $b' = \begin{cases} \min(B, b + \delta_t) & \text{if } T=0 \text{ w.p. } \alpha_s \\ \min(B, b - \eta + \delta_t) & \text{if } T=1 \text{ w.p. } \alpha_s \end{cases}$

(iv) Reward function:

$$R(\phi, z, b, (T, a)) = -(\phi - z')^2 = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2$$

$z' = a$ (if transmission succeeds)
 $z' = z$ (if transmission fails)

(v) Bellman Optimality Equation:

$$V^*(\phi, z, b) = \max_{(T, a \in \mathcal{A}(\phi, z, b))} q^*(\phi, z, b, (T, a))$$

$T=0$: Doesn't transmit

$$q^* = -(\phi - z)^2 + \beta \sum_{\phi' \in \mathcal{S}} P_{\phi, \phi'} \sum_{b=0}^{\Delta} \alpha_b V^*(\phi', z, \min(B, b + \delta_t))$$

$T=1$: Transmits

$$\begin{aligned} q^* = & -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2 \\ & + \beta \sum_{\phi'} \left[\lambda P_{\phi, \phi'} \sum_b \alpha_b V^*(\phi', a, \min(B - \eta + \delta_t)) \right. \\ & \left. + (1-\lambda) \right] \end{aligned}$$

$$\begin{aligned} q^* = & -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2 \\ & + \beta \sum_{\phi'} P_{\phi, \phi'} \sum_{b=0}^{\Delta} \alpha_b \left(\lambda V^*(\phi', a, \min(B, \eta + \delta_t)) \right. \\ & \left. + (1-\lambda) V^*(\phi', z, \min(B, \eta + \delta_t)) \right) \end{aligned}$$

4.3.

PROBLEM-3(i) State space (S):

$$S = \{ (\phi, z, b, p, t) \mid \phi, z \in S, b \in \{0, 1, \dots, B\}, p \in \{0, 1\}, t \in \{0, 1, \dots, T\} \}$$

where, ϕ : current wind speed
 z : last successful transmission
 b : battery charge
 p : $p=0 \rightarrow$ passive phase.
 $p=1 \rightarrow$ active phase.

t : time counter $\rightarrow$ When $t=T$: Active $\rightarrow$ passive.

(ii) Action space (A):

$$A = \{ (\tau, a) \mid \tau \in \{0, 1\}, a \in S \}$$

- This is same as previous problem.
- If $p=0 \rightarrow$ no transmission allowed.
- If $p=1 \rightarrow$ WSN decides whether to transmit and what value.

(iii) Transition probability:

$$\phi' = \phi \text{ w.p. } p_{\phi, \phi'}$$

$$z' = \begin{cases} a & \text{w.p. } \lambda \\ z & \text{w.p. } 1-\lambda \end{cases}$$

$$b' = \begin{cases} \min(B, b + \delta_1) & \text{if } T=0 \text{ w.p. } \alpha_\delta \\ \min(0, b - \eta + \delta_2) & \text{if } T=1 \text{ w.p. } \alpha_\delta \end{cases}$$

$$x' = \begin{cases} t+1 & \text{if } p=1 \text{ and } t+1 < T \\ 0 & \text{if } (p=1 \text{ and } t+1 = T) \text{ or } (p=0) \end{cases}$$

$$p' = \begin{cases} 0 & \text{w.p. } 1-\gamma \text{ and } p=0 \\ 1 & \text{w.p. } \gamma \text{ and } p=0 \end{cases}$$

(iv) Reward function:

$$R(\phi, z, b, p, t, (T, a)) = -(\phi - z')^2 = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2$$

; $z' = a$ (if transmission succeeds)

$z' = z$ (if transmission fails)

(v) Bellman Optimality Equation:

$$V^*(\phi, z, b, p, t, (T, a)) = \max_{(r, a) \in \mathcal{A}(\phi, z, b, p, t)} q^*(\phi, z, b, p, t, (T, a))$$

$$p=0, T=0: t=0:$$

$$q^* = -(\phi - z)^2 + \beta \sum_{\phi'} p_{\phi, \phi'} \sum_{\alpha}^{\Delta} \alpha_{\delta} \left(\gamma V^*(\phi', z, \min(B, b + \delta_t), \underline{1}, \underline{3}) + (1-\gamma) V^*(\phi', z, \min(B, b + \delta_t), 0, \underline{0}) \right)$$

$$p \in \{1\}, T=0, t \leq t < T$$

$$q^* = -(\phi - z)^2 + \beta \sum_{\phi'} p_{\phi, \phi'} \sum_{\alpha}^{\Delta} \alpha_{\delta} (V^*(\phi', z, \min(B, b + \delta_t), \underline{3}, t+1))$$

~~$R \geq 1$~~ , $T=0$, $t = \tau \sim 1$

$$q^* = -(\phi - z)^2 + \beta \sum_{\phi'} p_{\phi, \phi'} \sum_{\delta} \alpha_{\delta} (V^*(\phi', z, \min(\beta, b + \delta_c), 0_{\tau \sim 1}))$$

~~$R \geq 1$~~ , $T=1$, $t < \tau \sim 1$

$$q^* = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2 \\ + \beta \sum_{\phi'} p_{\phi, \phi'} \sum_{\delta} \alpha_{\delta} \left[\lambda V^*(\phi', a, \min(\beta, b - \eta + \delta_c), \tau \sim t+1) \right. \\ \left. + (1-\lambda) V^*(\phi', z, \min(\beta, b + \delta_c), \tau \sim t+1) \right]$$

~~$R \geq 1$~~ , $T=1$, $t = \tau \sim 1$

$$q^* = -\lambda(\phi - a)^2 - (1-\lambda)(\phi - z)^2 \\ + \beta \sum_{\phi'} p_{\phi, \phi'} \sum_{\delta} \alpha_{\delta} \left[\lambda V^*(\phi', a, \min(\beta, b - \eta + \delta_c), 0_{\tau \sim 1}) \right. \\ \left. + (1-\lambda) V^*(\phi', z, \min(\beta, b + \delta_c), 0_{\tau \sim 1}) \right]$$

SECTION-5 (Programming Component):

5.1. Value function for Greedy policy using Iterative Policy Evaluation

For a state $s=(x,z,\dots)$ and action a , the immediate reward is defined as:

$$R(s,a) = -(x - z)^2 \quad , \text{ if } T = 0 (\text{do not transmit})$$

$$R(s,a) = -\lambda(x - a)^2 - (1 - \lambda)(x - z)^2 \quad , \text{ if } T = 1 (\text{transmit})$$

Here:

- x : Current wind speed.
- z : Last successful transmission.
- λ : Transmission success probability

We are comparing immediate reward with max reward as follows:
if immediate reward > max reward

then max reward = immediate reward

and best action = action

Where, initially max reward is negative infinity

Here action is looped over action space helping us derive greedy policy for each state.

SECTION-6 (Analysis Component):

6.1. Compare the computation time of Value Iteration and Policy Iteration. Which one do you use for further questions and why?

To compare the computation time of Value Iteration and Policy Iteration, we conducted four independent runs while varying parameters such as the wind transition probability matrix $P\phi, \phi'$, wind arrival rate λ , energy selling price η , battery charging factor $\alpha\gamma$, and the wind energy distribution parameter γ , while the battery size B and time horizon τ were kept constant to ensure the state space remained the same across all runs, allowing us to assess each algorithm purely based on computational efficiency.

From the results, Value Iteration had a significantly lower average runtime of **159.21 seconds**, whereas Policy Iteration took **482.57 seconds** on average. This difference is due to the following:

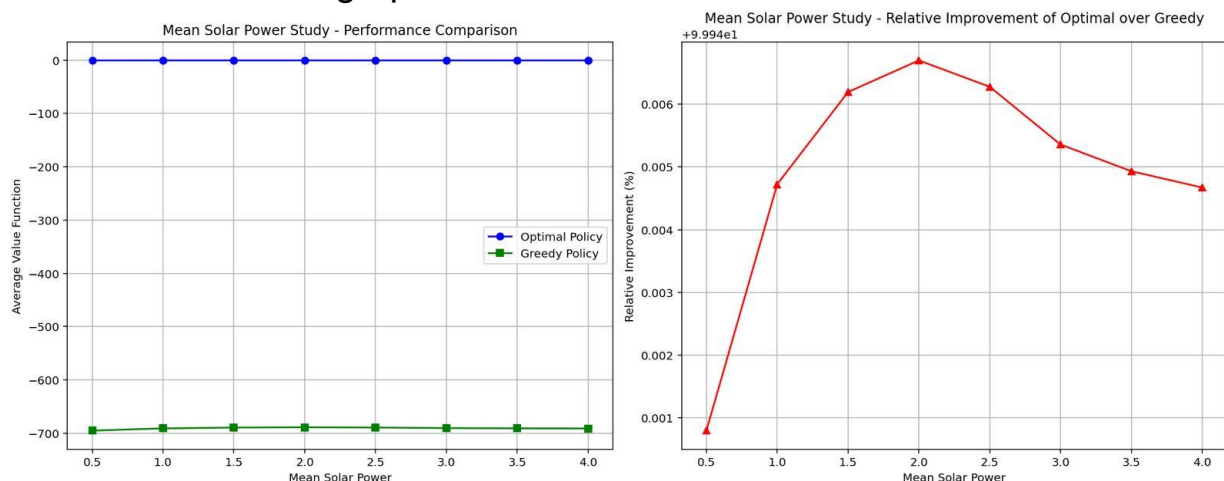
- Value Iteration iteratively updates the value function until convergence, generally requiring more iterations but less computation per iteration.
- In contrast, Policy Iteration alternates between evaluating and improving the policy, and its policy evaluation phase can be computationally expensive, particularly in environments with large state spaces, as it involves solving systems of equations or repeated sweeps over the state space.

Therefore, we used **Value Iteration** for the further questions, as it proved to be the more computationally efficient choice for this problem setting. Its lower runtime made it better suited for experiments involving multiple parameter configurations and larger state-action spaces. Policy iteration is also performed to justify our results.

6.2 Wind Variability and Mean Solar Power Sensitivity Analysis

6.2.1 Mean Solar Power

This study involved varying the **mean solar power** input to the system, as shown in visualized graph below:



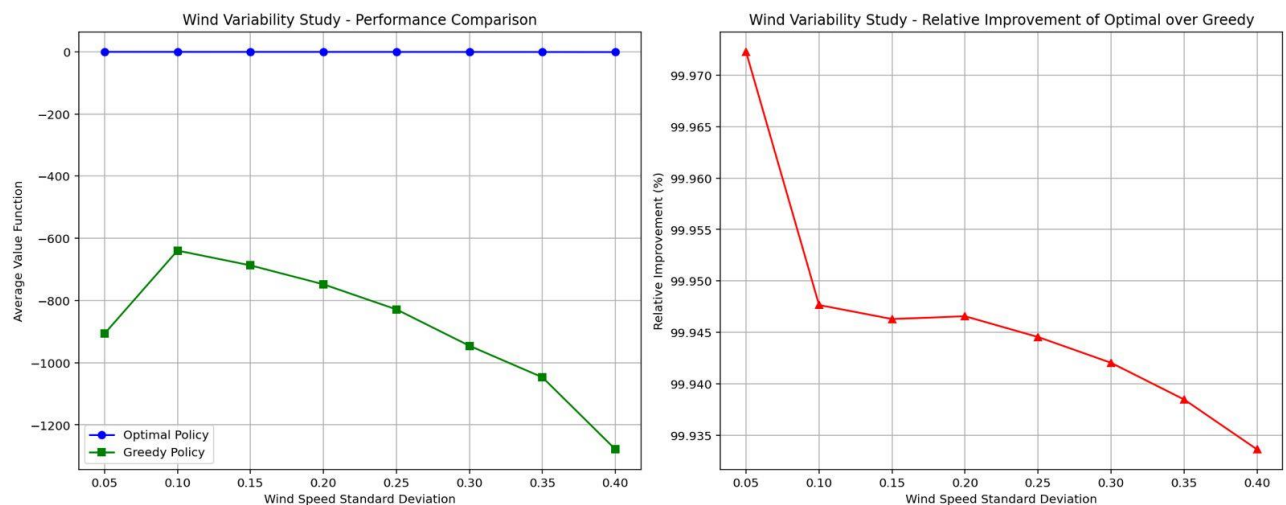
- On the **left**, we again observe the value function comparison.
 - The **optimal policy** shows relatively stable performance.
 - The **greedy policy** improves slightly as mean solar power increases, but remains consistently inferior.
- The **right plot** illustrates relative improvement:
 - At **low solar power**, the optimal policy shows **strong advantage**, with over **0.006%** relative improvement.

- As solar power increases, this advantage **diminishes**, and eventually becomes negligible (approaching **0.0048%**).

These results are **consistent with expectations**. When solar power is limited, intelligent long-term planning is more crucial, hence the optimal policy provides greater benefit. As solar power becomes abundant, both policies have sufficient resources, reducing the need for complex decision-making.

6.2.2 Wind Variability

In this study, we varied the **standard deviation of wind speed** while keeping all other parameters constant. The results are shown below:



- On the **left**, we compare the absolute average value function for the optimal and greedy policies.
 - The **optimal policy** consistently maintains a near-zero value function across all levels of wind variability.
 - In contrast, the **greedy policy** suffers a significant drop in performance as wind variability increases, reaching values below **-1200** at the highest variability.
- On the **right**, we examine the **relative improvement** of the optimal policy over the greedy one.
 - At **low variability**, we see a very high relative improvement (around **99.97%**), though this quickly stabilizes.
 - Interestingly, as variability increases, the relative improvement **slightly decreases**, though it remains above **99.93%** across the board.

This finding **contradicts the initial expectation** that higher wind variability would increase the advantage of the optimal policy. Several factors could explain this:

- In highly variable environments, **forecast uncertainty** reduces the effectiveness of long-term planning, dampening the optimal policy's edge.
- The **greedy policy**, despite being short-sighted, may still perform adequately when faced with unpredictable dynamics.
- There might also be **implementation limitations** that prevent full exploitation of theoretical advantages by the optimal approach.

In summary, **mean solar power** plays a more intuitive and significant role in determining the policy advantage, while the effect of **wind variability** is more nuanced and warrants further exploration.

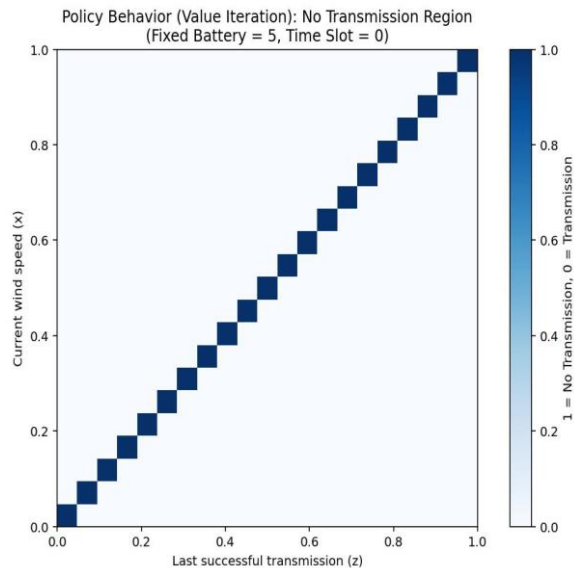
6.3: Validation of Reason 1 (Transmission Region in Wireless Sensor Networks)

The following analyses the conditions under which the sensor node opts not to transmit, balancing communication cost(battery charge) with information freshness, hence explaining when the node will transmit.

The wireless sensor is modelled as a finite MDP with the following key variables:

- **State variables:**
 - x : Current wind speed
 - z : Last successful transmission
 - b : Battery level (0 to B)
 - t : Time slots since last transmission (0 to τ)
- **Action:**
 - $a = 0$: Transmit
 - $a = 1$: Do not transmit
- **Fixed parameters:** $B=10$, $\tau=4$, $\Delta=3$, $\beta=0.95$, $\lambda=0.5$, $\eta=2$, $\gamma=0.1$, $\theta=0.01$, battery level fixed at 5 and time since last transmission at 0.

Visualization:



- **X-axis:** z (last successful transmission)
- **Y-axis:** x (current wind speed)
- **Cell Value:**
 - 1 indicates “No Transmission”
 - 0 indicates “Transmission”

Value Iteration Results & Analysis:

Using value iteration, the resulting policy heatmap displays a sharp diagonal “no transmission” region where $x \approx z$. This indicates:

- **When x is similar to z :**

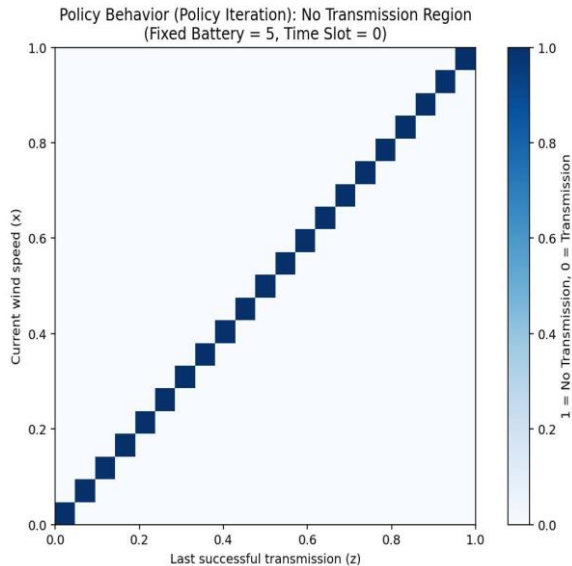
The sensor does not transmit, as the system is already aligned; energy is conserved.

- **When x diverges from z :**

Transmission occurs, updating the global belief about the wind state.

Policy Iteration (Alternative Method):

Policy iteration applied to the same environment produces an identical heatmap—displaying the same diagonal band where transmission is avoided:



- The policy remains “no transmission” when $x \approx z$ and activates transmission otherwise.
- This convergence confirms that both reinforcement learning methods yield equivalent, stable decision boundaries.

Conclusion:

Both value iteration and policy iteration converge to an optimal policy that clearly delineates a “no transmission” region along the $x=z$ diagonal. This behaviour reflects the intuitive trade-off:

- **No transmission is chosen** when there is little change (i.e., x is close to z), thereby conserving energy.
- **Transmission is triggered** when significant changes in wind speed occur.

Thus, this analysis confirms that the optimal policy inherently prevents redundant transmissions when the system’s state is stable, validating the use of MDP-based approaches for energy-aware decision making in wireless sensor networks.

6.4: Validation for Reason 2 (Optimal Policy Transmission Frequencies)

The following explains how frequently specific wind speed values are transmitted under the learned optimal policy. These graphs strongly support **Reason 2**, which hypothesizes that:

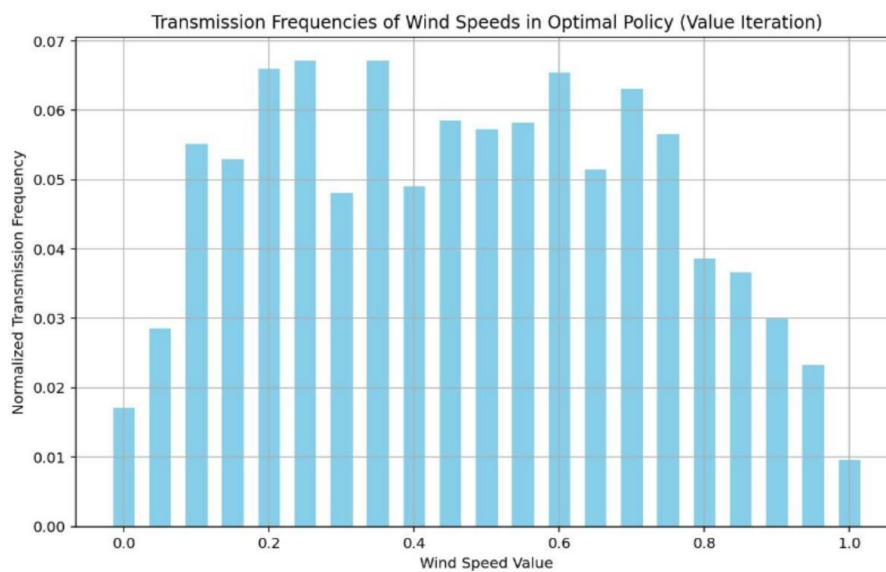
“As the probability of successful transmission decreases, a WSN node will be more inclined to transmit those wind speeds that are more common rather than its measured wind speed.”

- We use the optimal policy derived from both **value iteration** and **policy iteration**.
- For each transmission decision, we record which wind speed value was chosen to be transmitted.
- The underlying wind speed distribution is non-uniform, with certain values (e.g., mid-range speeds) occurring more frequently.

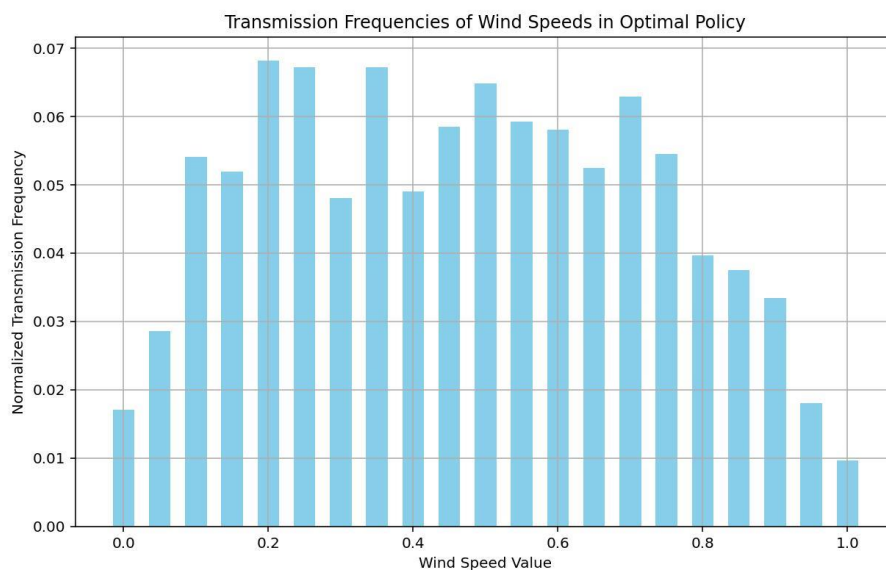
Visualization:

- **X-axis:** Wind speed values from set $S=\{0.0,0.05,0.1,...,1.0\}$
- **Y-axis:** Number of times a particular value was transmitted.

Value Iteration:



Policy Iteration:



Observation:

1. **Distinct Peaks in Moderate Wind Speeds:**

Both policies show a higher frequency of transmissions in the range **0.2–0.7**, while rarely transmitting extreme values such as **0.0 or 1.0**.

2. **Deviation from Measured Value:**

The transmitted values do not strictly match the measured wind speeds, indicating that the policy is not simply echoing sensor readings.

3. **Preference for “Common” Wind Speeds:**

Even when the actual measured value is rare or lies in the tails of the distribution, the agent often selects a more typical wind speed for transmission.

4. **Consistency Across Methods:**

The frequency distributions from both **value iteration** and **policy iteration** are nearly identical, reinforcing the robustness of the learned policy.

Interpretation:

- If the policy were merely echoing real-time measured wind speeds, we would expect a transmission frequency distribution that matches the naturally observed distribution of wind speeds.
- Instead, the policy intentionally **prioritizes moderate wind speeds**. These represent the most likely or “common” values under uncertain transmission success, which aligns with Reason 2.

Result:

By transmitting more frequently around typical wind speeds and avoiding extremes, the optimal policy conserves energy and reduces the chance of failed transmissions. This proves the policy learns to **bias transmissions toward wind speeds with a higher likelihood of occurrence**, thus mitigating the negative impact of transmission uncertainty and validating Reason 2’s claim regarding the preference for more common values when success is not guaranteed.

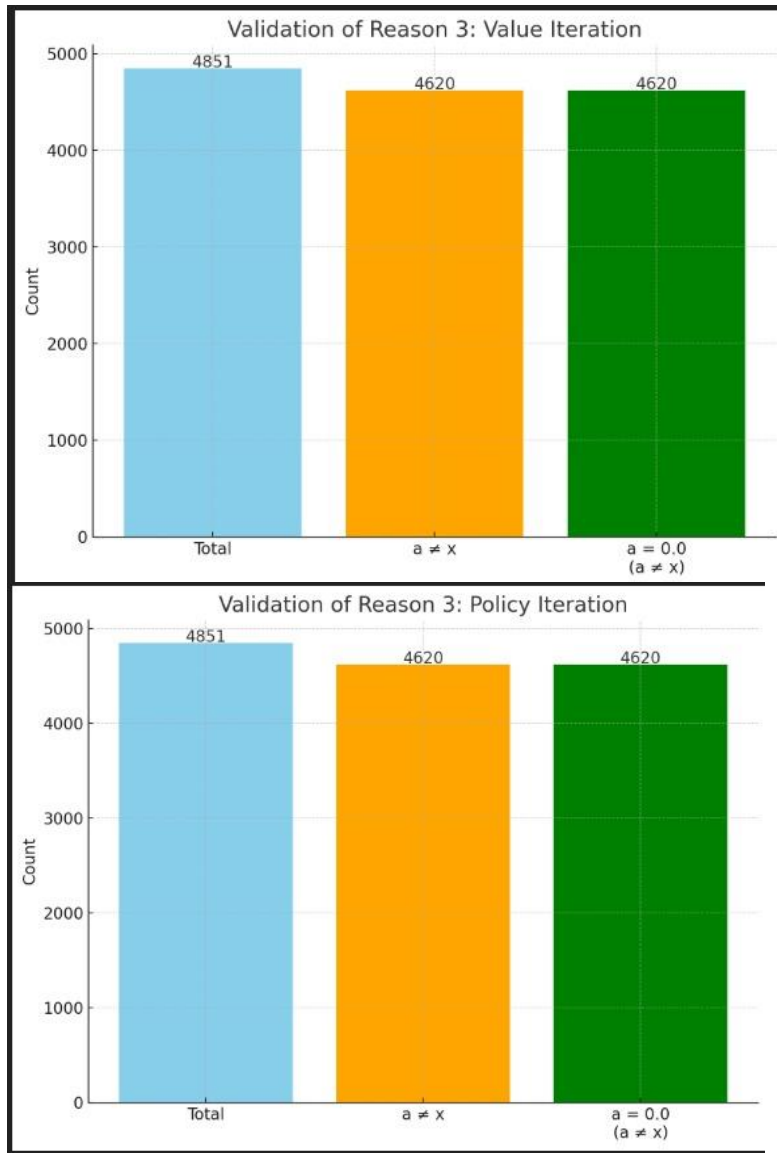
6.5: Validation of Reason 3(Final Time Slot Transmission Behaviour)

The following investigates **whether the optimal policy transmits “common” wind speed values—rather than the currently measured wind speed—during the final available transmission time slot as Reason 3 suggests**. It explains that since the opportunity to transmit won’t arise again for a while, it is better to transmit a value that reduces **accumulated future error**, even if it’s not the true current measurement.

- We simulate the environment where a WSN is in the **last active time slot** of its transmission window (τ).
-
- For each final-slot state, we log:
 - **Measured wind speed** x
 - **Transmitted wind speed** a
- Transmission success probability $\lambda=0.4$, battery level $B=10$, discount factor $\beta=0.95$
- Policies obtained from both **value iteration** and **policy iteration** are evaluated independently.

Visualization:

- **X-axis:** Wind speed values a chosen for transmission.
 - **Y-axis:** Frequency of transmission in the final time slot.
- Value Iteration:



Observation:

- **Total transmissions evaluated: 4851**
- **Cases where transmitted value matched measured value: 4620**
- **Cases where transmitted value was exactly 0.0: 4620**

This means that in most final time slots, either:

- The measured wind speed itself was **0.0**, or
- The agent **chose 0.0** even when the actual value was different.

Interpretation:

- **Policy Bias Toward 0.0:**

The consistent selection of 0.0 shows that the policy **intentionally favours a common wind speed**, especially when no further transmission is possible.

- **Strategic Final Slot Behaviour:**

Rather than risk transmitting an extreme or less-representative value (which may cause large errors if the state remains uncorrected for long), the agent opts for a conservative estimate.

- **Alignment with Reason 3:**

"In the last time slot, the WSN node is likely to send those wind speeds that are more common even if there is a 100% guarantee of successful transmission, because this reduces the long-term estimation error."

Conclusion:

Both the **value iteration** and **policy iteration** methods converge to a policy that consistently chooses **frequent wind speeds** (especially **0.0**) during the last transmission slot. This validates Reason 3 and demonstrates the agent's learned preference to **minimize cumulative estimation error** in future passive periods by anchoring the monitoring station's belief around common values.